s.no	Rate of a reaction	Rate constant of a reaction
2	It is measured as decrease in the concentration of the reactants or increase in the concentration of products.	It is equal to the rate of reaction, when the concentration of each of the reactants is unity
3	It depends on the initial concentration of reactants.	It does not depend on the initial concentration of reactants.

7.4 Molecularity:

Kinetic studies involve not only measurement of a rate of reaction but also proposal of a reasonable reaction mechanism. Each and every single step in a reaction mechanism is called an elementary reaction.

An elementary step is characterized by its molecularity. The total number of reactant species that are involved in an elementary step is called molecularity of that particular step. Let us recall the hydrolysis of t-butyl bromide studied in XI standard. Since the rate determining elementary step involves only t-butyl bromide, the reaction is called a Unimolecular Nucleophilic substitution (S_N^1) reaction.

Let us understand the elementary reactions by considering another reaction, the decomposition of hydrogen peroxide catalysed by I^-.

$$2H_2O_2(aq) \longrightarrow 2H_2O(l) + O_2(g)$$

It is experimentally found that the reaction is first order with respect to both H_2O_2 and I^-,which indicates that I^- is also involved in the reaction. The mechanism involves the following steps.

Step:1

$$H_2O_2(aq) + I^-(aq) \longrightarrow H_2O(l) + OI^-(aq)$$

Step : 2

$$H_2O_2(aq) + OI^-(aq) \rightarrow H_2O(l) + I^-(aq) + O_2(g)$$

Overall reaction is

$$2H_2O_2(aq) \longrightarrow 2H_2O(l) + O_2(g)$$

These two reactions are elementary reactions. Adding eqn (1) and (2) gives the overall reaction. Step 1 is the rate determining step, since it involves both H_2O_2 and I^-, the overall reaction is bimolecular.

Differences between order and molecularity:

s.no	Order of a reaction	Molecularity of a reaction
1	It is the sum of the powers of concentration terms involved in the experimentally determined rate law.	It is the total number of reactant species that are involved in an elementary step.
2	It can be zero (or) fractional (or) integer	It is always a whole number, cannot be zero or a fractional number.
3	It is assigned for a overall reaction.	It is assigned for each elementary step of mechanism.

XII U7 kinetics - Jerald Folder.indd 210 22-12-2022 19:53:54